

The Clock That Never Ticks

A machine-checked path from the dilogarithm
to quantum speed limits in Lean 4

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Abstract

Mathlib’s source contains a comment referring to the dilogarithm Li_2 — a function the library does not have. We fill that hole and follow the questions where they lead. The result is a `sorry`-free Lean 4 development ($\sim 1\,500$ lines over Mathlib) containing, to the best of our knowledge, several firsts in any proof assistant: the real dilogarithm with Euler’s reflection identity, Landen’s transformation, the duplication formula, and the golden-ratio ladder $\text{Li}_2(1/\varphi^2) = \pi^2/15 - \ln^2 \varphi$, $\text{Li}_2(1/\varphi) = \pi^2/10 - \ln^2 \varphi$ (derived from a 3×3 linear system — no five-term relation needed); the Rogers value $L(1/\varphi^2) = \pi^2/15$, equivalently the Lee–Yang effective central charge $c_{\text{eff}} = 2/5$, the simplest thermodynamic–Bethe–ansatz dilogarithm identity; the Clausen function Cl_2 and Catalan’s constant $G = \text{Cl}_2(\pi/2)$; the Fejér–Jackson inequality $\sum_{k=1}^M \sin(k\theta)/k > 0$ on $(0, \pi)$; the positivity $\text{Cl}_2(\theta) \geq \sin(\theta)/2$ by an asymptotics-free Abel summation; and the Margolus–Levitin and (an L^1 form of the) Mandelstam–Tamm quantum speed limits. The pieces assemble into the ending the title promises: the quantum state with populations $\propto 1/n^2$ on equally spaced energy levels has *infinite* mean energy and *infinite* energy variance — both textbook speed limits say nothing — and yet *never* reaches a state orthogonal to itself, because its autocorrelation is $\frac{6}{\pi^2} \text{Li}_2(e^{-i\theta})$ and the dilogarithm has no zero on the unit circle. A clock with an unbounded energy budget that never ticks. We are exact about the boundary: every identity here is classical (Euler, Landen, Clausen, Fejér, Jackson, Mandelstam–Tamm, Margolus–Levitin); the contribution is the machine-checked development and its assembly. Every named theorem depends only on the three standard axioms.

How to read this paper. It is written as the story of six questions, in the order they actually arose; each section opens with its question and ends by handing the reader the next one. Figure 1 is the map. If you read only one thing, read Theorem 4 and Figure 4; the Lean inventory is Table 1, and reproduction takes three commands (Appendix A).

1 The question that starts it: why does the library cite a function it does not have?

The file `Mathlib/Analysis/SpecialFunctions/Integrals/LogTrigonometric.lean` of Mathlib — the mathematical library of the Lean 4 proof assistant, on which a substantial part of contemporary

*Formalized with AI assistance (Claude, Anthropic); the mathematics and all claims are the author’s responsibility. The methodology and the exact boundary of the contribution are discussed in Section 7.

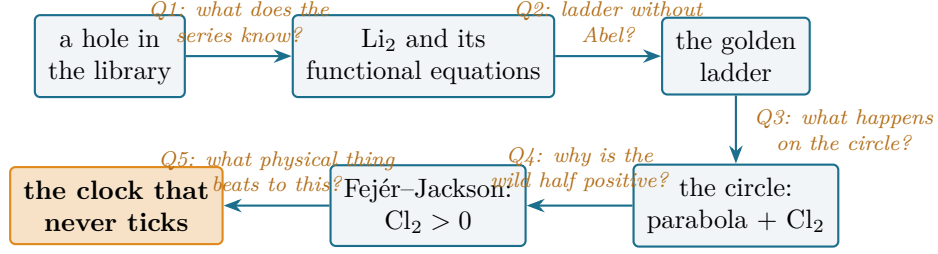


Figure 1: The journey, question by question. Every node is a block of `sorry`-free Lean; every arrow is the question that produced the next block.

formalized mathematics is built — describes one of its integrals by referring to the dilogarithm

$$\text{Li}_2(x) = \sum_{n \geq 1} \frac{x^n}{n^2}.$$

The function itself is absent. Not just from Mathlib: as far as we could determine, the dilogarithm, the Clausen function, Catalan’s constant, the Fejér–Jackson inequality, and every quantum speed limit are absent from *every* proof assistant (Lean, Rocq/Coq, Isabelle); the existing quantum libraries (CoqQ, SQIR, Isabelle Marries Dirac) formalize circuits, not dynamical bounds.¹

How does a function this central stay unformalized for a decade of Mathlib? Because it sits in an odd blind spot: too special to be infrastructure, too analytic to be algebra, and guarded — as we will see — by exactly one genuinely delicate limit. The genesis of this project was an audit: a fifty-digit numerical verification of the dilogarithm identities in a recent preprint of Alha [14]. Auditing someone else’s dilogarithms sent us to the library for a reference implementation; the library sent us back empty-handed. We decided to fill the hole, and to keep asking the next question until the questions stopped. They stopped five sections later, inside a quantum mechanics problem. This paper is the travelogue.

2 Q1: what does the series know about its own derivative?

History bead. The function is Euler’s (1768); the reflection identity below is his. Landen’s transformation is from 1780. Everything in this section is classical (Lewin [1], Zagier [2]); the Lean is `Dilog/Basic.lean`.

A power series differentiates termwise, and the dilogarithm’s derivative collapses to elementary functions:

$$\text{Li}'_2(x) = \sum_{n \geq 1} \frac{x^{n-1}}{n} = -\frac{\ln(1-x)}{x} \quad (|x| < 1, x \neq 0).$$

In Lean this single line is the analytic core of the whole development — term-by-term differentiation under a geometric majorant (`hasDerivAt_tsum_of_isPreconnected`), then the Mercator series to identify the sum. Once the derivative is known, functional equations become mechanical:

¹Library and literature checks performed June 2026: Loogle and `grep` over Mathlib (zero `quantum` declarations; no `dilog/polylog/Clausen`, and Catalan *numbers* only); web searches over the Archive of Formal Proofs and the Rocq/Coq ecosystems. Absence of evidence, with that caveat.

Recipe: a functional equation in three moves

To prove a combination F of dilogarithms and logarithms is constant on an interval: (1) differentiate termwise and check $F' = 0$ by chain/product rules plus `field_simp`; `ring`; (2) conclude F is constant by mean-value machinery (`constant_of_has_deriv_right_zero` on subintervals); (3) evaluate the constant as a *boundary limit*. Step (3) is where the one delicate limit of the subject lives: $\ln x \cdot \ln(1-x) \rightarrow 0$ at the endpoint, tamed as $(\ln(1-x)/x) \cdot (x \ln x) \rightarrow (-1) \cdot 0$ via the slope characterization of the derivative and `tendsto_log_mul_rpow_nhdsGT_zero`. Continuity of Li_2 up to the boundary is the Weierstrass M -test (`continuousOn_tsum`).

Theorem 1 (`Li2_add_Li2_one_sub`, `Li2_landen`, `Li2_sq`, `Li2_one_half`). *On the stated domains,*

$$\begin{aligned} \text{Li}_2(x) + \text{Li}_2(1-x) &= \frac{\pi^2}{6} - \ln x \ln(1-x) & x \in (0, 1) \\ \text{Li}_2(x) + \text{Li}_2\left(\frac{x}{x-1}\right) &= -\frac{1}{2} \ln^2(1-x) & x \in (0, \tfrac{1}{2}) \\ \text{Li}_2(x^2) &= 2(\text{Li}_2(x) + \text{Li}_2(-x)) & |x| \leq 1 \\ \text{Li}_2(\tfrac{1}{2}) &= \frac{\pi^2}{12} - \frac{\ln^2 2}{2}. \end{aligned}$$

Two honesty notes, both visible in the Lean statements. The Landen domain stops at $x = \frac{1}{2}$: past it, the argument $x/(x-1)$ leaves the disc of convergence, and a power-series definition does not analytically continue — we do not pretend it does. And the duplication formula needs no calculus at all: it is a pure rearrangement of the series by even and odd indices (`tsum_even_add_odd`), the formal mirror of how Euler would have proved it.

The turn. Three functional equations in hand, the classical showpieces of the subject are the golden-ratio evaluations — and every textbook reaches them through Abel’s five-term relation, a two-variable identity we have *not* formalized. Are we stuck?

3 Q2: can the golden ladder be climbed without Abel’s relation?

History bead. The values are Landen’s (1780); the two-term sum is from Rogers’ 1907 memoir [4]. Their role as a central charge belongs to the thermodynamic Bethe ansatz of the 1990s [9, 10, 11].

The answer is yes, and the key is two arithmetic accidents of the golden ratio $\varphi = (1 + \sqrt{5})/2$, both one-liners from $\varphi^2 = \varphi + 1$:

$$\frac{1}{\varphi} + \frac{1}{\varphi^2} = 1 \quad \text{and} \quad \frac{1/\varphi^2}{1/\varphi^2 - 1} = -\frac{1}{\varphi}.$$

The first says $1/\varphi$ and $1/\varphi^2$ are a *reflection pair*; the second says *Landen maps* $1/\varphi^2$ to $-1/\varphi$ — and note $1/\varphi^2 \approx 0.382 < \frac{1}{2}$, inside our proven Landen domain. Reflection at $1/\varphi^2$, Landen at $1/\varphi^2$, duplication at $1/\varphi$: three linear equations, three unknowns, and `linarith` closes the system.

$$\text{Li}_2\left(\frac{1}{\varphi^2}\right) = \frac{\pi^2}{15} - \ln^2 \varphi, \quad \text{Li}_2\left(\frac{1}{\varphi}\right) = \frac{\pi^2}{10} - \ln^2 \varphi, \quad \text{Li}_2\left(-\frac{1}{\varphi}\right) = \frac{\ln^2 \varphi}{2} - \frac{\pi^2}{15}.$$

No five-term relation — a small but pleasant discovery of the formalization process. In the Rogers normalization $L(x) = \text{Li}_2(x) + \frac{1}{2} \ln x \ln(1-x)$ the $\ln^2$ defects cancel (`rogersL_inv_goldenRatio_sq`, `rogersL_gold_sum`):

$$L\left(\frac{1}{\varphi^2}\right) = \frac{\pi^2}{15}, \quad L\left(\frac{1}{\varphi}\right) + L\left(\frac{1}{\varphi^2}\right) = \frac{\pi^2}{6} = L(1),$$

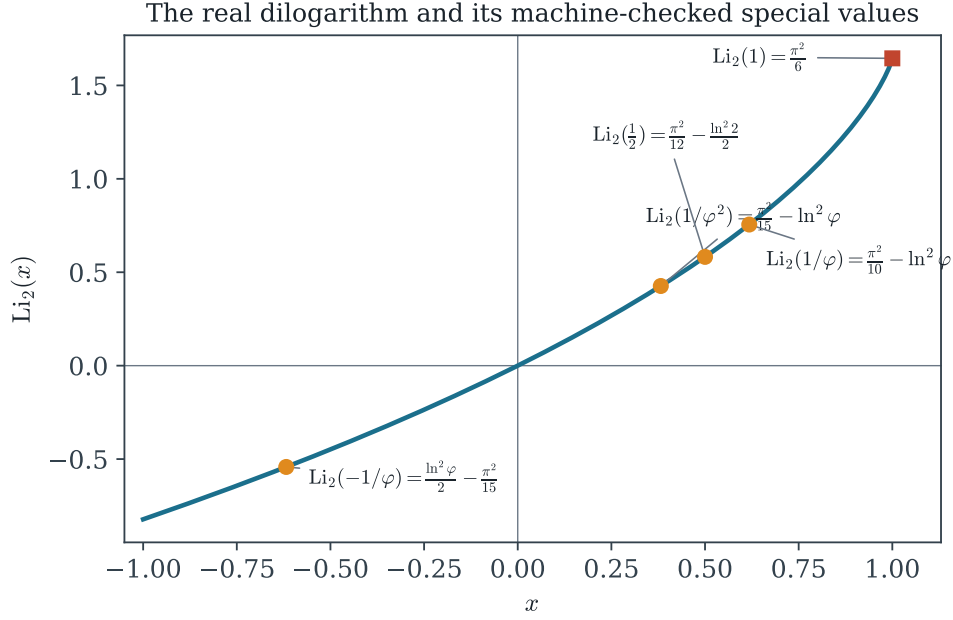


Figure 2: The real dilogarithm. Every marked value is a **sorry-free** Lean theorem; the generating script asserts each against 40-digit numerics before drawing.

and the first, normalized, is the effective central charge of the Lee–Yang thermodynamic Bethe ansatz, $c_{\text{eff}} = L(1/\varphi^2)/L(1) = 2/5$ — to our knowledge the first TBA dilogarithm identity verified in a proof assistant. Figure 2 shows the curve with its certified values.

The turn. The ladder spends the function’s life *inside* the interval of convergence. The natural next territory is the boundary itself. What does the series do on the unit circle?

4 Q3: what happens on the circle?

History bead. Clausen introduced his function in 1832 [3]; Catalan’s constant is from his 1865 memoir, and its irrationality is, famously, still open. The cosine series below is the Fourier expansion of the Bernoulli polynomial B_2 , which Mathlib already owns (`hasSum_one_div_nat_pow_mul_cos`); the Lean of this section is `Dilog/Clausen.lean`.

On the circle the dilogarithm splits into a tame half and a wild half:

$$\begin{aligned} \text{Li}_2(e^{i\theta}) &= \underbrace{\sum_{n \geq 1} \frac{\cos n\theta}{n^2}}_{= \frac{\pi^2}{6} - \frac{\pi\theta}{2} + \frac{\theta^2}{4}} + i \underbrace{\sum_{n \geq 1} \frac{\sin n\theta}{n^2}}_{= \text{Cl}_2(\theta)} \quad (\theta \in [0, 2\pi]). \end{aligned}$$

The real part is an *elementary parabola* (`hasSum_cos_div_sq`, from Mathlib’s Bernoulli–Fourier machinery at $k = 1$). The imaginary part is not elementary at all: it is the **Clausen function**, a genuinely new special function — the reason this section exists. At the quarter circle it produces a celebrity:

$$G = \text{Cl}_2\left(\frac{\pi}{2}\right) = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2} = 0.9159\dots,$$

Catalan’s constant, absent from every proof assistant until now (`catalanConst`, `Cl2_pi_div_two`; the proof is the same even/odd rearrangement as the duplication formula — even-indexed sines vanish, odd-indexed give the alternating series). Basic symmetries come for free: $\text{Cl}_2(0) = \text{Cl}_2(\pi) = 0$, Cl_2 is odd, and $\text{Cl}_2(2\pi - \theta) = -\text{Cl}_2(\theta)$.

The turn. Plot the wild half (Figure 3) and one suspicion dominates everything: between 0 and π the function looks *positive*. Two values and a hunch are not a theorem. Why is it positive? That question is older than it looks: it is Fejér’s, from 1910.

5 Q4: why is the wild half positive?

History bead. Fejér raised and partially proved the positivity of the sine partial sums in 1910; Jackson’s proof is from 1911 [5, 6]. It is the grandfather of the theory of positive trigonometric sums (Viotoris, Askey). The Lean is `Dilog/FejerJackson.lean`.

Theorem 2 (`fejersum_pos`). *For every $M \geq 1$ and $\theta \in (0, \pi)$: $\sum_{k=1}^M \frac{\sin k\theta}{k} > 0$.*

Hand-checkable case. $M = 2$: $\sin \theta + \frac{1}{2} \sin 2\theta = \sin \theta (1 + \cos \theta) > 0$ on $(0, \pi)$. The point of the formalization is that the same certainty now covers every M .

The proof is the classical interior-minimum induction, with two touches worth recording. First, the Dirichlet kernel is handled *multiplicatively* — no division, no removable singularity:

$$2 \sin \frac{\theta}{2} \sum_{k=1}^M \cos k\theta = \sin((M + \frac{1}{2})\theta) - \sin \frac{\theta}{2},$$

a telescope of $\sin((k + \frac{1}{2})\theta)$ via the product-to-sum identity (`Finset.sum_range_sub`). Second, at an interior local minimum the critical equation $\sin((M + \frac{1}{2})\theta_0) = \sin(\theta_0/2)$ *factors* (`Real.sin_sub_sin`) as $\sin(M\theta_0/2) \cos((M+1)\theta_0/2) = 0$, and its two branches force

$$\sin(M\theta_0) = 0 \quad \text{or} \quad \sin(M\theta_0) = \sin \theta_0 > 0 :$$

either way the last term of the sum is nonnegative at the minimum, so the minimum value exceeds the previous partial sum — positive by induction. Boundary values vanish; compactness closes.

From Fejér–Jackson to the Clausen function is an Abel summation, and here the formalization found a small simplification: *no asymptotics are needed*. Writing the $(N+1)$ -term partial sum of the Cl_2 series in terms of the Fejér–Jackson sums S_M ,

$$\sum_{n=1}^{N+1} \frac{\sin n\theta}{n^2} = \frac{S_{N+1}(\theta)}{N+1} + \sum_{k=1}^N S_k(\theta) \left(\frac{1}{k} - \frac{1}{k+1} \right),$$

every piece on the right is positive, and the $k = 1$ summand alone equals $\sin(\theta)/2$. Every partial sum is therefore $\geq \sin(\theta)/2$ — no harmonic-number bounds, no tail estimates — and the limit inherits the bound (`Cl2_pos`):

$$\text{Cl}_2(\theta) \geq \frac{\sin \theta}{2} > 0 \quad (0 < \theta < \pi).$$

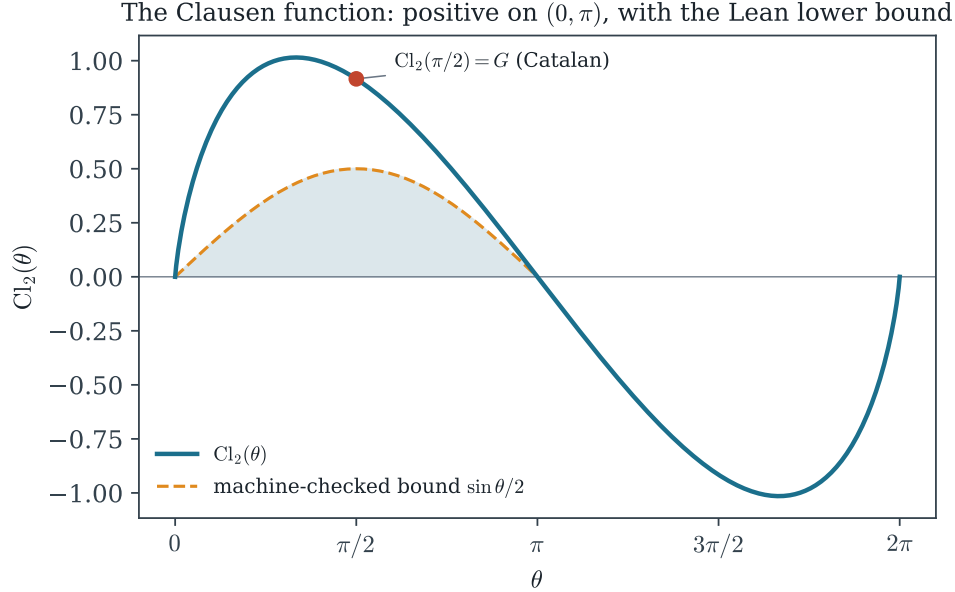


Figure 3: The Clausen function on $[0, 2\pi]$ with the machine-checked lower bound $\sin(\theta)/2$ (shaded) on $(0, \pi)$, and Catalan’s constant at the quarter circle. The generating script asserts the bound on a 400-point grid at 30 digits.

The turn — the big one. Step back and look at what we are holding. A sum $\sum p_n e^{-in\theta}$ with positive weights $p_n \propto 1/n^2$, whose imaginary part we have just proven never vanishes on $(0, \pi)$. A physicist reads that expression instantly: it is the *autocorrelation of a quantum state* — the inner product between a state and its own time evolution, for populations p_n on equally spaced energy levels. The circle has been measuring *time* all along, and our positivity theorem is a statement about a quantum system that cannot reach an orthogonal state. How fast *can* any quantum system reach one?

6 Q5: what physical thing beats to this function?

History bead. Mandelstam and Tamm bounded the orthogonalization time by the energy *spread* in 1945 [7]; Margolus and Levitin bounded it by the mean energy in 1998 [8]. Together they are the textbook “quantum speed limits”: how many distinguishable states can a system visit per unit time, on a given energy budget. The Lean is QSL/.

A state $|\psi_0\rangle = \sum_n c_n |E_n\rangle$ evolves with autocorrelation $S(t) = \sum_n p_n e^{-iE_n t}$, $p_n = |c_n|^2$, and reaches an orthogonal state at time τ iff $S(\tau) = 0$. This is a statement about a scalar function of a discrete nonnegative spectral measure, and that is how we formalize it — the Hilbert space is bookkeeping. Orthogonality enters as two real conditions ($\operatorname{Re} S = \operatorname{Im} S = 0$); Margolus–Levitin needs no complex numbers at all.

Theorem 3 (`margolus_levitin`, `mandelstam_tamm_L1`; $\hbar = 1$). *If $S(\tau) = 0$ with $\tau \geq 0$, then*

$$\pi \leq 2 \langle E \rangle \tau \quad \text{and} \quad 1 \leq D_1 \tau, \quad D_1 = \sum_n p_n |E_n - \bar{E}|.$$

The heart of Margolus–Levitin is one inequality of independent interest, $\cos x \geq 1 - \frac{2}{\pi}(x + \sin x)$ for $x \geq 0$ (equality at 0 and π), which we prove by the same unimodal pattern as Fejér–Jackson:

rise then fall, both endpoints zero, split at $x_0 = 2 \arctan(2/\pi)$. The same proof shape carried three theorems in this paper — that is not a coincidence but a transferable technique. Since $D_1 \leq \Delta E$ by Cauchy–Schwarz, the second bound implies the textbook weak Mandelstam–Tamm $\tau \geq 1/\Delta E$; the sharp constant $\pi/2$ needs the geodesic argument and is left as a road (§8).

Now the dénouement. Take populations $p_n = \frac{6}{\pi^2 n^2}$ on levels $E_n = n$: the *weight-2 zeta state*. Its autocorrelation *is* our protagonist,

$$S(\theta) = \frac{6}{\pi^2} \text{Li}_2(e^{-i\theta}) = \frac{6}{\pi^2} \left(\underbrace{\frac{\pi^2}{6} - \frac{\pi\theta}{2} + \frac{\theta^2}{4}}_{\text{the parabola of §4}} - i \underbrace{\text{Cl}_2(\theta)}_{\text{positive, by §5}} \right).$$

Its mean energy $\sum_n p_n n$ *diverges*. So does its energy variance. Both bounds of Theorem 3 are vacuously satisfied: on this state, the speed limits of quantum mechanics predict nothing at all. One might guess such an energy-unbounded state orthogonalizes instantly. The truth is the opposite extreme:

Theorem 4 (`zetaState_never_orthogonal`). *For every $\theta \in (0, 2\pi)$, the real and imaginary parts of the weight-2 zeta-state autocorrelation are not both zero: the state never reaches an orthogonal state.*

Proof (as in Lean). On $(0, \pi)$ the imaginary part is $-\text{Cl}_2(\theta) \neq 0$ by the Fejér–Jackson bound. At $\theta = \pi$ the real part is the parabola value $-\pi^2/12 \neq 0$ (it is $-\eta(2)$). On $(\pi, 2\pi)$, the reflection $\text{Cl}_2(2\pi - \theta) = -\text{Cl}_2(\theta)$ hands the sign back to the first case. $\square$

In fact $|S| \geq \frac{6}{\pi^2} \cdot \frac{\pi^2}{12} = \frac{1}{2}$ on the whole circle: the state never gets even *halfway* to distinguishability (Figure 4). A clock with an infinite energy budget whose hand never completes a tick — because the budget is spent on a heavy spectral tail rather than on coherent separation. Energy is necessary for evolution speed (Margolus–Levitin); this is the machine-checked counterpoint that it is nowhere near sufficient.

Remark 1 (Which clocks do tick: a closing observation). *For monotone non-increasing populations on equally spaced levels, ticking (orthogonalization) means a zero of the power series $\sum_n p_n z^n$ on the unit circle — and the Eneström–Kakeya theorem (1893/1912) [12] localizes such zeros: they occur only in the equality cases, where the decrements $p_{n-1} - p_n$ concentrate on an arithmetic progression, placing the zeros at roots of unity. The Margolus–Levitin–sharp state $(|0\rangle + |2E\rangle)/\sqrt{2}$ is exactly such an equality case ($\frac{1}{2}(1+z)$, zero at $z = -1$). We have not found this dictionary — root location of positive-coefficient polynomials $\leftrightarrow$ orthogonalizability of monotone quantum states — in either literature; we record it as an observation and a road.*

7 The boundary of the contribution

Everything here that is mathematics is classical: Euler (1768), Landen (1780), Clausen (1832), Rogers (1907), Fejér (1910), Jackson (1911), Mandelstam–Tamm (1945), Margolus–Levitin (1998); the TBA reading of the golden values is from the 1990s. We claim no new identity and no new inequality. The contributions are: (i) the machine-checked development itself — per our checks of June 2026, the first formalization in any proof assistant of each item listed in the abstract; (ii) three proof-engineering choices that may be reusable: the division-free Dirichlet telescope, the asymptotics-free Abel bound $\text{Cl}_2 \geq \sin(\theta)/2$, and the fully real (scalar spectral-measure) statements of the speed limits; (iii) the assembly of Theorem 4, whose physical reading we have not seen stated,

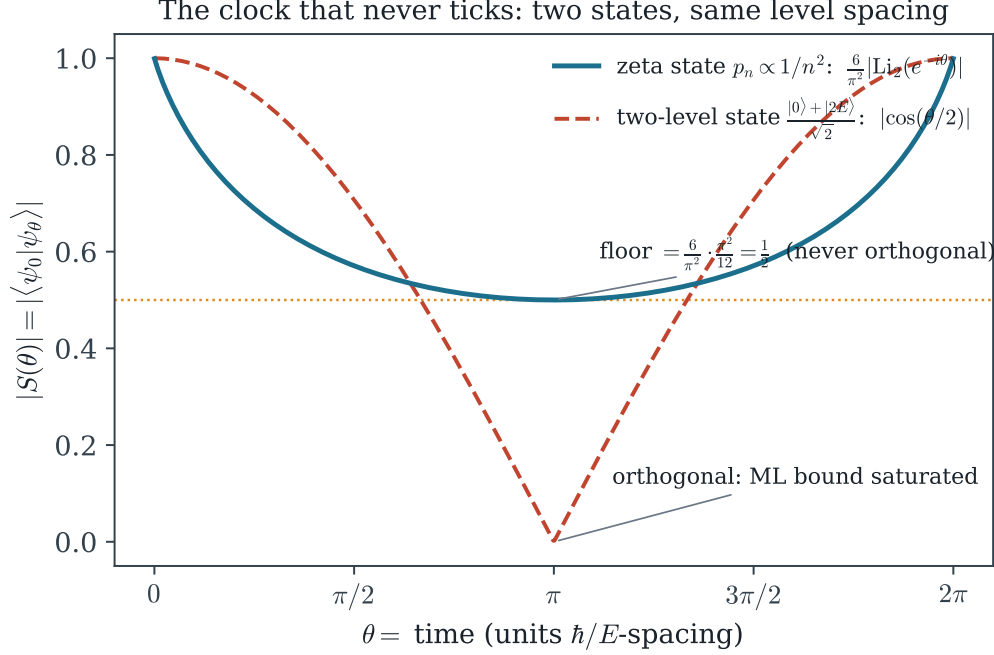


Figure 4: Two states with the same level spacing. The two-level state $(|0\rangle + |2E\rangle)/\sqrt{2}$ (dashed) saturates Margolus–Levitin and strikes orthogonality at $\theta = \pi$. The zeta state (solid), with *infinite* mean energy, has $|S| \geq \frac{1}{2}$: the clock that never ticks. The script asserts the floor equals $\eta(2) = \pi^2/12$ at $\theta = \pi$.

though we would not be surprised to find it folklore among polylogarithm or QSL specialists; and (iv) the observation of Remark 1.

On methodology: the formalization was carried out in collaboration with an AI assistant (Claude, Anthropic), with every step gated by the Lean kernel, every identity checked independently at 20–50 digits *before* formalization, and every named theorem verified by `#print axioms` to depend only on `propext`, `Classical.choice`, `Quot.sound`. The mathematics, the mistakes, and the claims remain the author’s responsibility.

8 Open roads

1. **Abel’s five-term relation** — the capstone of weight 2; with it, every two-variable dilogarithm identity follows. Our reflection/Landen machinery is the warm-up.
2. **The complex dilogarithm** and the Bloch–Wigner function; with them, the 2D dimer entropy (G/π — Catalan again) and hyperbolic volumes come into reach.
3. **Sharp Mandelstam–Tamm** ($\pi/2\Delta E$, the geodesic argument) and the arbitrary-fidelity Margolus–Levitin bound — conjectured in 2003, proven only in 2023 [13]: a fresh theorem to formalize.
4. **Eneström–Kakeya**, both as a formalization target and as the dictionary of Remark 1.
5. **Positive trigonometric sums beyond Fejér–Jackson** (Vietoris) — the toolkit is in place.

A Reproducing

```
git clone https://github.com/karlesmarin/godsil-gutman-lean
cd godsil-gutman-lean && lake exe cache get
lake build Dilog QSL
```

Lean toolchain v4.30.0-rc2, Mathlib pinned at 701fb6e9. The figures regenerate with `python figures/dilog_fig*.py`; each script asserts its mathematical content against 20–40-digit numerics before drawing.

B Lean inventory

File	Contents	Lines
Dilog/Basic.lean	Li ₂ : series, derivative, reflection, Landen, duplication, Li ₂ ($\frac{1}{2}$), golden ladder, Rogers L, Lee–Yang	581
Dilog/Clausen.lean	Cl ₂ , Catalan’s G , Bernoulli parabola, $\text{Cl}_2 \geq \sin \theta/2$, $\text{Cl}_2(2\pi - \theta) = -\text{Cl}_2(\theta)$, Theorem 4	332
Dilog/FejerJackson.lean	Fejér–Jackson inequality	189
QSL/Basic.lean	Margolus–Levitin (+ the cosine inequality)	238
QSL/MandelstamTamm.lean	Mandelstam–Tamm, L^1 form	134
		1 474

Table 1: The development. All theorems `sorry-free`; axioms: `propext`, `Classical.choice`, `Quot.sound` throughout.

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